

HONOLULU INTL, HI, USA

WMO: 911820

Lat: 21.324N

Lon: 157.929W

Elev: 2

StdP: 101.30

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22521

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	18.4	10.9	8.1	21.9	12.0	8.7	21.8	11.0	24.5	9.9	24.4	1.9	320	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	32.0	23.4	31.5	23.2	31.0	22.9	25.2	29.2	24.7	28.8	24.2	28.5	7.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	27.3	23.4	18.2	27.1	22.9	17.6	26.8	76.7	29.2	74.6	28.7	72.6	28.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	9.1	8.5	DB	15.2	32.7	1.2	1.0	14.4	33.4	13.7	34.0	13.0	34.6	12.1	35.3	(4)
			WB	13.0	25.8	1.1	0.9	12.3	26.5	11.6	27.0	11.0	27.5	10.3	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	23.2	23.3	23.7	24.7	25.5	26.7	27.4	27.8	27.5	26.8	25.5	24.1
	DBStd	2.02	1.37	1.34	1.30	1.26	1.28	0.94	0.94	0.91	1.01	1.11	1.25	1.32
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5664	408	372	424	441	482	502	540	551	524	521	463	436
	CDD18.3	2623	150	138	166	191	223	252	282	292	274	263	213	178
	CDH23.3	20246	671	597	776	1105	1644	2221	2746	2984	2708	2376	1500	918
	CDH26.7	4961	42	37	73	182	362	614	838	967	854	635	269	87
Wind	WSAvg	4.6	3.8	4.1	4.4	4.9	4.6	5.2	5.4	5.3	4.6	4.4	4.3	4.1
Precipitation	PrecAvg	477	69	52	51	29	26	10	14	16	21	46	60	77
	PrecMax	1086	326	240	213	227	184	62	69	194	114	283	374	439
	PrecMin	93	1	2	0	2	1	1	1	0	1	2	1	0
	PrecStd	218	74	56	47	37	34	14	15	29	20	51	73	86
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	28.7	29.1	30.3	31.2	31.9	32.2	32.8	32.7	31.9	30.8	29.6
		MCWB	21.5	21.8	20.9	22.1	22.5	22.6	23.6	23.7	23.9	23.4	23.2	23.1
	2%	DB	27.7	27.7	28.2	29.3	30.2	31.0	31.5	32.0	31.8	31.1	29.8	28.4
		MCWB	21.1	21.1	20.9	21.4	22.2	22.5	23.1	23.4	23.6	23.2	22.6	21.6
	5%	DB	27.0	27.0	27.5	28.5	29.4	30.3	31.0	31.4	31.2	30.4	29.0	27.6
		MCWB	20.9	20.8	20.9	21.2	21.9	22.3	22.8	23.1	23.3	22.9	22.3	21.1
	10%	DB	26.3	26.3	26.8	27.8	28.7	29.7	30.4	30.7	30.5	29.7	28.3	26.9
		MCWB	20.7	20.5	20.7	21.0	21.5	22.0	22.6	22.7	23.0	22.6	22.0	21.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	23.8	23.5	24.0	24.7	24.7	25.2	25.7	26.1	25.4	25.0	24.4
		MCDB	26.4	26.7	26.6	27.7	28.3	28.9	29.4	29.5	30.1	29.2	28.1	27.5
	2%	WB	23.0	22.8	22.7	22.9	23.6	23.9	24.5	24.8	25.3	24.6	24.1	23.3
		MCDB	25.5	25.9	26.0	27.1	27.9	28.5	29.1	29.3	29.3	28.6	27.4	26.2
	5%	WB	22.4	22.1	22.1	22.3	22.9	23.2	23.9	24.2	24.6	24.1	23.5	22.6
		MCDB	25.3	25.6	25.5	26.6	27.7	28.3	28.8	29.1	28.9	28.2	27.1	25.8
	10%	WB	21.7	21.4	21.5	21.7	22.3	22.7	23.4	23.7	24.1	23.6	22.9	21.9
		MCDB	25.0	25.1	25.1	26.1	27.2	28.0	28.5	28.8	28.6	28.0	26.8	25.5
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.5	6.4	6.4	6.6	6.4	6.4	6.5	6.7	6.3	6.0	6.1
		MCDBR	7.1	6.8	7.0	6.9	7.0	6.8	6.6	7.0	7.0	6.8	6.4	6.5
		MCWBR	3.2	3.0	2.9	2.5	2.5	2.3	2.2	2.2	2.3	2.3	2.5	2.8
	5% WB	MCDBR	6.2	6.3	6.3	6.5	6.6	6.5	6.4	6.6	6.6	6.3	5.9	5.9
		MCWBR	3.5	3.4	3.3	2.8	3.0	2.6	2.5	2.5	2.7	2.7	2.9	3.2
Clear-Sky Solar Irradiance	taub	0.363	0.370	0.379	0.391	0.384	0.371	0.370	0.368	0.362	0.377	0.376	0.359	
	taud	2.575	2.544	2.530	2.481	2.532	2.575	2.590	2.593	2.615	2.576	2.571	2.610	
	Ebn at Noon	912	924	927	913	909	913	915	921	930	905	891	905	
	Edh at Noon	92	101	106	113	106	101	99	99	96	96	92	87	
All-Sky Solar Radiation	RadAvg	4.45	5.00	5.73	6.47	6.92	7.24	7.13	6.85	6.40	5.39	4.54	4.11	
	RadStd	0.19	0.35	0.60	0.37	0.40	0.21	0.15	0.27	0.19	0.17	0.12	0.29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+68	+68

Nomenclature: See separate page